

YEAR 7 MATHEMATICS · CHAPTER 10

Analysing data

10.01 Interpreting graphs**10.03** Dot plots**10.05** Mean and mode**10.07** Analysing dot plots and stem-and-leaf plots**10.02** Misleading graphs**10.04** Stem-and-leaf plots**10.06** Median and range**10.08** Comparing data

Name

Class

Teacher

How to use this booklet

- Each lesson starts with **learning intentions** and **success criteria**. Write the **key terms** as your teacher explains them, and use the **notes** box.
- Work through the levels in order: Level 1 , then Level 2, then Level 3. Extension questions stretch you further.
- Some questions ask you to **draw** a diagram, graph or construction. Use the space given, a ruler and a sharp pencil.
- At the end of each lesson, **show off your skill**: choose one question, then tear off the ticket and hand it in.
- Missed a lesson? Each lesson ends with a **summary** that has a worked example for every level. Start there, then try the questions.

WE DO

Grey boxes with a dark border. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box and write your answer.

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In this chapter you will

- interpret picture, column, sector and line graphs
- identify misleading graphs
- read and draw dot plots and stem-and-leaf plots
- find the mean, mode, median and range
- analyse data shown in plots
- compare two sets of data

NSW Mathematics K–10 Syllabus (2022), Stage 4 Statistics: MA4-DAT-C-01 classifies and displays data using a variety of graphical representations; MA4-DAT-C-02 analyses simple datasets using measures of centre, range and shape of the data.



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
10.01

Interpreting graphs

Date _____

Learning intentions

We are learning to:

- read and interpret picture, sector, column and line graphs

Success criteria

I can:

- ☐ read the title, axes, key and scale first
- ☐ read values from a graph
- ☐ answer questions and explain trends

Key terms Write the definition as your teacher explains it.

Key

Scale

Sector graph

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

Set A graph: what is the most popular destination?

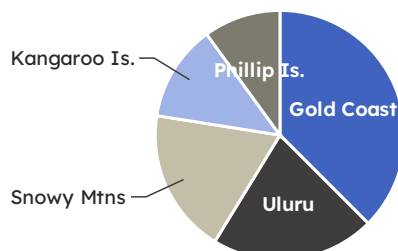
Example 2

Set A graph: which destination did 17 people prefer?

YOU DO

On your own: Set A. Use the sector graph.

Favourite holiday destinations (80 people)

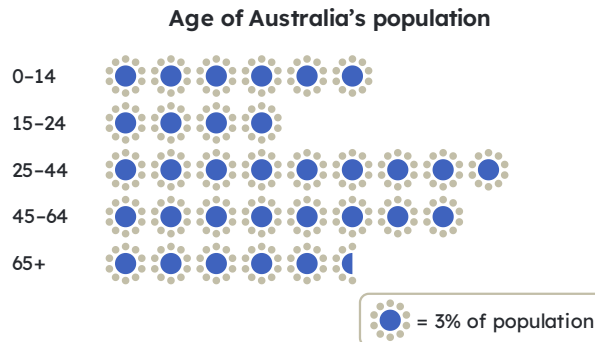


1 Most popular destination?

2 Which did 17 people prefer?

3 Estimate how many chose the Gold Coast.

4 True or false: fewer than 10 chose Phillip Island.

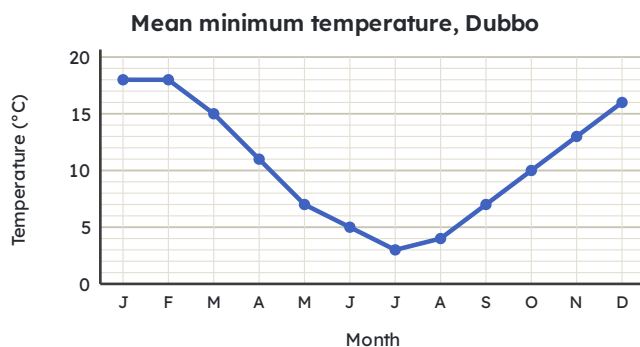
YOU DO On your own: Set B. Use the picture graph.

- 1 Smallest age group?
- 2 Percentage aged 15-24?
- 3 Percentage aged 65+?
- 4 Why might 25-44 be the largest group?

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 Set B graph: which age group has the smallest percentage?

Example 4 Set B graph: what percentage are aged 15-24?

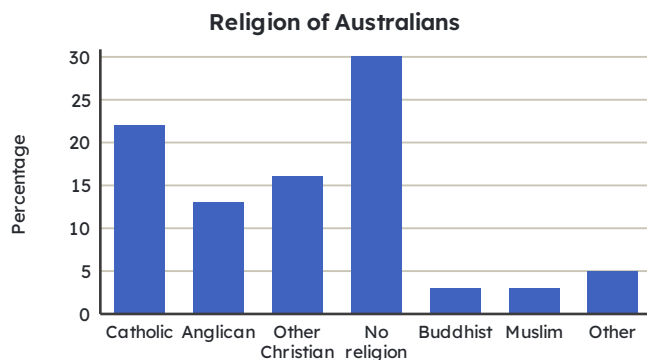
YOU DO On your own: Set C. Use the line graph.

- 1 Highest months and temperature?
- 2 Two months with the same temperature?
- 3 Which month has 10°C?
- 4 Coldest month?
- 5 Between which 2 months is the smallest drop?
- 6 Describe the trend from January to July.



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Set C graph: which months have the highest mean minimum? What is it?
Example 6 Set D graph: why might this graph have been different 20 years ago?
YOU DO On your own: Set D. Use the column graph.**1** What percentage are Catholic?
2 Which religion is followed by 13%?
3 What is the most common response?
4 How might this graph have been different 20 years ago?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

In the sector graph, which destination was least popular?

Level 2

In the line graph, what is the mean minimum in April?

Level 3

In the column graph, how many more % have no religion than are Catholic?

Extension · Stretch yourself

Worked example

In the sector graph, what angle is the Gold Coast sector?

1. Gold Coast is 30 out of 80 people
2. $\frac{30}{80} \times 360^\circ$

Answer: 135°

YOU DO

On your own: read the worked example, then try these.

- E1** Find the angle of the Uluru sector (1 d.p.).

- E2** Draw a divided bar graph 16 cm long for the holiday data. How long is each part?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Read the title, labels and key.
 3. Read values carefully with a ruler.

2. Check the scale on each axis.
4. Describe trends: increasing, decreasing, highest, lowest.

Level 1

Most popular (sector)

1. Largest sector

Answer: Gold Coast

Level 2

15–24 (picture graph)

1. 4 symbols $\times$ 3%

Answer: 12%

Level 3

Highest mean minimum

1. Highest points

Answer: Jan and Feb,
 18°C

Extension

Gold Coast angle

1. $30/80 \times 360$

Answer: 135°

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LESSON
10.02

Misleading graphs

Date _____

Learning intentions

We are learning to:

- identify features that make a graph misleading

Success criteria

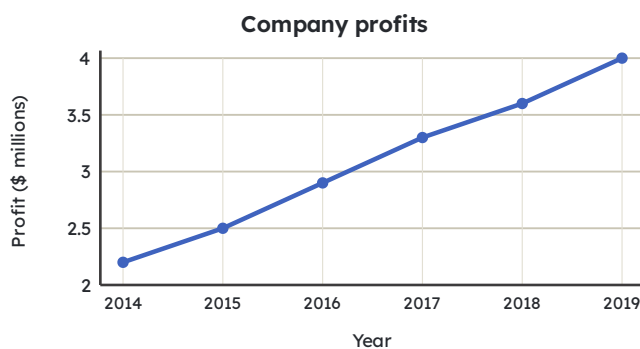
I can:

- ☐ check whether the vertical axis starts at zero
- ☐ check the scale is even and the graph has labels
- ☐ redraw a misleading graph correctly

Key terms Write the definition as your teacher explains it.**Misleading****Broken axis**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Set A graph: what is incorrect about it?**Example 2** Set A graph: what misleading impression does it give?**YOU DO** On your own: Set A. Use the line graph.

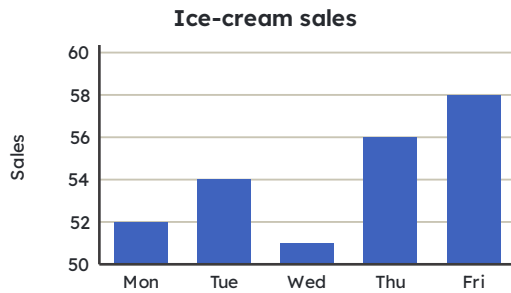
1 What is incorrect about the graph?

2 What impression does it give?

3 What was the profit in 2014?

4 By how much did profit grow from 2014 to 2019?

YOU DO On your own: Set B. Use the column graph.



- 1 Where does the vertical axis start?
- 2 Wednesday's sales?
- 3 Friday's sales?
- 4 Does Friday have double Wednesday's sales?

Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Set B graph: Thursday sales look double Wednesday's. Is that true?

Example 4 Set B graph: what are the actual sales on Wednesday and Thursday?

YOU DO On your own: Set C. Redraw the company profits graph correctly, starting the axis at 0.

- 1 Profits 2014–2019: 2.2, 2.5, 2.9, 3.3, 3.6, 4.0 (\$ millions)

Redraw here



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Name 3 other ways a graph can mislead.
Example 6 Why would a company want a misleading graph?
YOU DO On your own: Set D. Think it through.

- 1 A graph uses pictures of bags of money; the 2019 bag is twice as tall and twice as wide as the 2014 bag. Why is this misleading?

- 3 A column graph has a scale 0, 10, 20, 50, 100. What is wrong?

- 2 A graph has no title and no labels on the axes. Why is this a problem?

- 4 Find a graph in a newspaper or online. Is it misleading? How?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Give one reason a graph might be misleading.

Level 2

A column graph's axis starts at 90. Why is that a problem?

Level 3

How should the company profits graph be fixed?

Extension · Stretch yourself

Worked example

Profit grew from \$2.2 million to \$4.0 million. By what percentage did it grow?

- Increase = 1.8 million
- $\frac{1.8}{2.2} \times 100\%$

Answer: $\approx 81.8\%$

YOU DO

On your own: read the worked example, then try these.

- E1** On the misleading graph, the 2019 point looks about 8 times higher than 2014. Explain why.

- E2** Draw a graph of the same data that makes growth look very small.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Check the vertical axis starts at 0 (or shows a break).
 2. Check the scale goes up in equal steps.
 3. Check there is a title and labelled axes.
 4. Watch for pictures that change in 2 directions.

Level 1

Profit graph

1. Axis starts at 2

Answer: growth looks bigger

Level 2

Sales Wed vs Thu

1. 51 vs 56

Answer: not double

Level 3

Uneven scale 0, 10, 20, 50

1. Steps not equal

Answer: misleading

Extension

Growth %

1. $1.8 \div 2.2$

Answer: $\approx 81.8\%$

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LESSON
10.03

Dot plots

Date _____

Learning intentions

We are learning to:

- read and draw dot plots

Success criteria

I can:

- ☐ read the frequency of each value
- ☐ find the mode and any outlier
- ☐ draw a dot plot from data

Key terms Write the definition as your teacher explains it.

Dot plot

Outlier

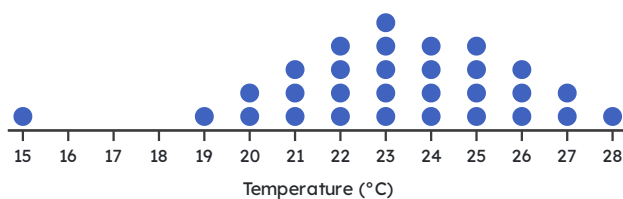
Mode

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Set A plot: what is the mode?**Example 2** Set A plot: what is the outlier?**YOU DO** On your own: Set A. Use the dot plot.

Maximum temperatures at Bega in April

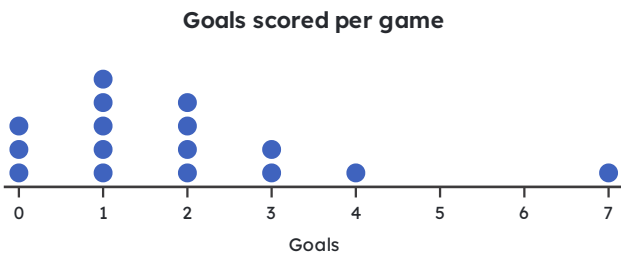


1 Mode?

2 Outlier?

3 Days at 25°C?

4 Fraction of days below 20°C?

YOU DO On your own: Set B. Use the dot plot.

1 How many games?

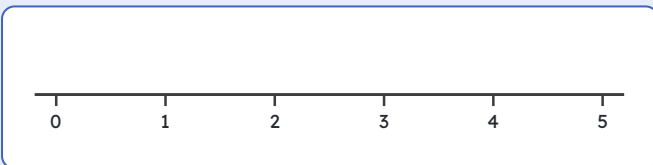
2 Mode?

3 Outlier?

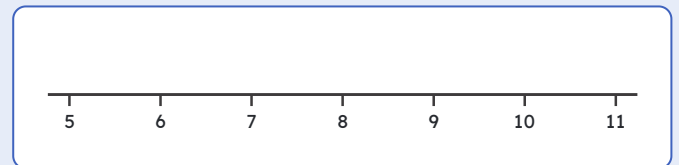
4 How many games had 2 or more goals?

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Set A plot: on how many days was it 25°C?**Example 4** Set A plot: on what fraction of days was it below 20°C?**YOU DO** On your own: Set C. Draw a dot plot for each set of data.

1 Number of pets: 0, 1, 1, 2, 0, 3, 1, 2, 1, 0, 4, 1



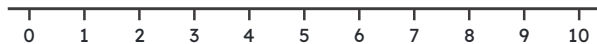
2 Shoe sizes: 7, 8, 8, 9, 7, 10, 8, 9, 8, 6, 8, 9



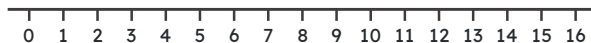


Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Draw a dot plot: 2, 3, 3, 4, 4, 4, 5, 6, 6, 9**Example 6** Describe the shape of the Set A plot.**YOU DO** On your own: Set D. Draw and analyse.

- 1 Minutes late: 0, 2, 1, 0, 3, 0, 15, 1, 2, 0, 1, 0. Draw a dot plot.



- 2 For Question 1, find the mode and the outlier. How does the outlier affect the data?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What is a dot plot?

Level 2

In Set B, what is the mode?

Level 3Draw a dot plot for 1, 2, 2, 3, 3, 3, 8.
Which value is the outlier?

Extension · Stretch yourself

Worked example

What percentage of April days at Bega were 25°C or more?

- Count: $25 (4) + 26 (3) + 27 (2) + 28 (1) = 10$
- $\frac{10}{30} \times 100\%$

Answer: 33.3%

YOU DO

On your own: read the worked example, then try these.

- E1** What percentage of games had no goals? (Set B)

- E2** Add a game with 3 goals to Set B. What is the new mode?

Lesson summary

Missed the lesson? Start here.

- Remember**
- Each dot is one piece of data.
 - Mode: the tallest column.
 - Outlier: far away from the rest.
 - Draw: a number line with an even scale, one dot per value.

Level 1

Mode of Bega plot

- Tallest column

Answer: 23°C

Level 2

Outlier

- Far from the rest

Answer: 15°C

Level 3

Below 20°C

- 2 days out of 30

Answer: $\frac{1}{15}$

Extension

25°C or more

- 10 of 30 days

Answer: 33.3%

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LESSON
10.04

Stem-and-leaf plots

Date _____

Learning intentions

We are learning to:

- read and draw ordered stem-and-leaf plots

Success criteria

I can:

- ☐ read values from a stem-and-leaf plot using the key
- ☐ order the leaves in each row
- ☐ draw a stem-and-leaf plot from data

Key terms Write the definition as your teacher explains it.

Stem

Leaf

Key

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Set A plot: how many players are there?**Example 2** Set A plot: what is the lowest mass?**YOU DO** On your own: Set A. Use the stem-and-leaf plot.

Masses of football players (kg), unordered

Stem	Leaf
7	9 5 8 1 6 9 4
8	2 5 0 0 8 0 3 2 4
9	0 8 4 2 2 0
10	5 6 4

Key: 7 | 9 = 79 kg

1 How many players?

2 Lowest mass?

3 Most common mass?

4 Highest mass?

YOU DO On your own: Set B. Use the stem-and-leaf plot.

Test scores	
Stem	Leaf
4	5
5	0 0 1 2
6	1 3 4 6 6 6 7 8
7	4 5

Key: 5 | 2 = 52

① How many scores?

② Lowest score?

③ Highest score?

④ Mode?

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Set A plot: which mass occurs most often?**Example 4** Set A plot: what is the highest mass?**YOU DO** On your own: Set C. Draw an ordered stem-and-leaf plot.

① 23, 35, 41, 28, 32, 45, 39, 27, 30, 44, 36, 21

Stem	Leaf
2	
3	
4	

Key: ____ | ____ = ____

② 67, 82, 75, 91, 78, 64, 88, 70, 73, 85, 79, 68

Stem	Leaf
6	
7	
8	
9	

Key: ____ | ____ = ____



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Draw an ordered stem-and-leaf plot for Set A.

Stem	Leaf
7	
8	
9	
10	

Key: ____ | ____ = _____

Example 6 Why do we order the leaves?

YOU DO On your own: Set D. Draw and analyse.**1** Order the Set A plot (masses of players).

Stem	Leaf
7	
8	
9	
10	

Key: ____ | ____ = _____

2 For the ordered plot, how many players weigh 90 kg or more? What fraction is that?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1In the key $6 | 3 = 63$, what is the stem?

Level 2Write the values in the row $4 | 0 \ 2 \ 7$.

Level 3

Draw an ordered stem-and-leaf plot for 12, 25, 17, 21, 14, 29.

Extension · Stretch yourself

Worked example

Write the data 3.4, 3.7, 4.1, 4.5, 4.5, 5.0, 5.2 as a stem-and-leaf plot.

1. Stems are the whole numbers, leaves the tenths
2. Key: $3 \mid 4 = 3.4$

Answer: $3 \mid 4 \ 7; 4 \mid 1 \ 5 \ 5; 5 \mid 0 \ 2$

YOU DO

On your own: read the worked example, then try these.

- E1** Draw a stem-and-leaf plot for 105, 112, 118, 121, 125, 130 (stems 10, 11, 12, 13).

- E2** What would a key of $2 \mid 5 = 0.25$ mean?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. The stem is the tens digit; the leaf is the ones digit.
 2. Always include a key, e.g. $7 \mid 9 = 79$.
 3. Order the leaves from smallest to largest.
 4. Line the leaves up in columns so the plot shows the shape.

Level 1

Players

1. Count the leaves

Answer: 25

Level 2

Lowest mass

1. First leaf of first stem

Answer: 71 kg

Level 3

Ordered plot

1. Sort each row

Answer: $7 \mid 1 \ 4 \ 5 \ 6 \ 8 \ 9 \ 9;$
 $8 \mid 0 \ 0 \ 0 \ 2 \ 2 \ 3 \ 4 \ 5 \ 8; 9 \mid 0$
 $0 \ 2 \ 2 \ 4 \ 8; 10 \mid 4 \ 5 \ 6$

Extension

Decimals

1. Stem = whole number

Answer: $3 \mid 4 \ 7; 4 \mid 1 \ 5 \ 5;$
 $5 \mid 0 \ 2$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.05

Mean and mode

Date _____

Learning intentions

We are learning to:

- find the mean and mode of a set of data

Success criteria

I can:

- ☐ find the mean: add the values, divide by how many
- ☐ find the mode: the most common value
- ☐ round the mean to 2 decimal places

Key terms Write the definition as your teacher explains it.

Mean

Mode

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

Find the mean of 4, 3, 2, 5, 6, 4, 4.

Example 2

Find the mode of 4, 3, 2, 5, 6, 4, 4.

YOU DO

On your own: Set A. Find the mode.

1 4, 3, 2, 5, 6, 4, 4

2 6, 12, 11, 12, 10, 6, 6, 10, 6

3 8, 4, 1, 1, 4, 1, 3, 6

4 3, 7, 7, 2, 9, 3, 7

5 10, 12, 15, 12, 10, 12

6 red, blue, red, green, blue, red

YOU DO On your own: Set B. Find the mean.

1 2, 4, 6, 8

4 5, 5, 5, 5

2 4, 3, 2, 5, 6, 4, 4

5 1, 2, 3, 4, 5, 6

3 10, 20, 30, 40, 50

6 7, 9, 11

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Find the mean of 6, 12, 11, 12, 10, 6, 6, 10, 6.**Example 4** Find the mode of 6, 12, 11, 12, 10, 6, 6, 10, 6.**YOU DO** On your own: Set C. Find the mean (2 d.p.) and mode.

1 6, 12, 11, 12, 10, 6, 6, 10, 6

3 12, 15, 11, 15, 17, 14

5 45, 52, 38, 52, 61, 49, 52

2 8, 4, 1, 1, 4, 1, 3, 6

4 3.5, 4.2, 3.5, 5.1

6 0, 0, 1, 2, 2, 2, 3, 5



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find the mean and mode of 8, 4, 1, 1, 4, 1, 3, 6.**Example 6** The mean of 4 numbers is 7. Three are 5, 8 and 9. Find the fourth.
YOU DO On your own: Set D. Problems.

- 1 The mean of 4 numbers is 7. Three are 5, 8 and 9. Find the fourth.

- 3 Write 5 numbers with a mean of 10 and a mode of 8.

- 2 Sam's test marks are 72, 85, 68 and 91. What must he score on the next test for a mean of 80?

- 4 Can a data set have more than one mode? Give an example.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the mode: 5, 8, 5, 3, 5, 8.

Level 2

Find the mean: 4, 6, 8, 10, 12.

Level 3

Find the mean (2 d.p.): 3, 7, 7, 9.

Extension · Stretch yourself

Worked example

A class of 25 has a mean mark of 64. A new student scores 90. Find the new mean.

1. Total = $25 \times 64 = 1600$
2. New total 1690, 26 students
3. $1690 \div 26$

Answer: 65

YOU DO

On your own: read the worked example, then try these.

- E1** The mean of 6 numbers is 12. One number, 20, is removed. Find the new mean.

- E2** Explain why the mean does not have to be one of the data values.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Mean = sum of values $\div$ number of values.
 2. Mode = the value that occurs most often.
 3. There can be more than one mode, or none.
 4. Round the mean to 2 decimal places.

Level 1

4, 3, 2, 5, 6, 4, 4

1. sum 28, 7 values

Answer: mean 4

Level 2

4, 3, 2, 5, 6, 4, 4

1. 4 occurs 3 times

Answer: mode 4

Level 3

8, 4, 1, 1, 4, 1, 3, 6

1. sum 28 $\div$ 8

Answer: mean 3.5,
mode 1

Extension

New student

1. $1690 \div 26$

Answer: 65

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.06

Median and range

Date _____

Learning intentions

We are learning to:

- find the median and range of a set of data

Success criteria

I can:

- ☐ order the data before finding the median
- ☐ find the middle value, or the mean of the two middle values
- ☐ find the range: highest – lowest

Key terms Write the definition as your teacher explains it.**Median****Range**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

Find the median of 4, 3, 2, 5, 6, 4, 4.

Example 2

Find the range of 4, 3, 2, 5, 6, 4, 4.

YOU DO

On your own: Set A. Find the range.

1 4, 3, 2, 5, 6, 4, 4

2 6, 12, 11, 12, 10, 6, 6, 10, 6

3 8, 4, 1, 1, 4, 1, 3, 6

4 15, 3, 22, 9

5 104, 97, 110, 99

6 -3, 5, 2, -1

YOU DO On your own: Set B. Find the median.

1 3, 5, 8, 9, 12

4 10, 7, 3, 9, 1

2 4, 3, 2, 5, 6, 4, 4

5 4, 4, 5, 7, 9, 10

3 2, 4, 6, 8

6 20, 15, 30, 25

Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Find the median of 6, 12, 11, 12, 10, 6, 6, 10, 6.

Example 4 Find the range of 6, 12, 11, 12, 10, 6, 6, 10, 6.

YOU DO On your own: Set C. Find the median and range.

1 6, 12, 11, 12, 10, 6, 6, 10, 6

3 12, 15, 11, 15, 17, 14

5 45, 52, 38, 52, 61, 49, 52

2 8, 4, 1, 1, 4, 1, 3, 6

4 3.5, 4.2, 3.5, 5.1

6 100, 2, 3, 4, 5

**Level 3 · Challenge**

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find the median and range of 8, 4, 1, 1, 4, 1, 3, 6.**Example 6** Which is affected more by an outlier: the median or the mean?**YOU DO** On your own: Set D. Problems.**1** Find the mean, median, mode and range of 2, 3, 3, 4, 5, 25.**3** Write 5 numbers with median 6 and range 10.**2** Which measure (mean or median) best describes the data in Question 1? Why?**4** The range of a set is 12 and the lowest value is 45. What is the highest?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO**Show off your skill**Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the median: 6, 2, 9, 4, 7.

Level 2

Find the range: 18, 5, 12, 30.

Level 3

Find the median: 3, 8, 10, 11.

Extension · Stretch yourself

Worked example

Five numbers have mean 6, median 5, mode 3 and range 8. Find them.

1. Mode 3 and median 5: 3, 3, 5, ?, ?
2. Sum = 30, so the last two add to 19
3. Range 8: largest = $3 + 8 = 11$, so the other is 8

Answer: 3, 3, 5, 8, 11

YOU DO

On your own: read the worked example, then try these.

- E1** Four numbers have median 7, mode 9 and range 6. Find them (hint: 9 appears twice).

- E2** Why might a company report the median salary instead of the mean?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Order the data first.
 2. Median = the middle value.
 3. Even number of values: median = mean of the middle two.
 4. Range = highest – lowest.

Level 1

4, 3, 2, 5, 6, 4, 4

1. Ordered: 2, 3, 4, 4, 4, 5, 6

Answer: median 4

Level 2

6, 12, 11, 12, 10, 6, 6, 10, 6

1. Ordered: 6, 6, 6, 6, 10, 10, 11, 12, 12

Answer: median 10

Level 3

8, 4, 1, 1, 4, 1, 3, 6

1. $8 - 1$

Answer: range 7

Extension

mean 6, median 5, mode 3, range 8

1. Build the list

Answer: 3, 3, 5, 8, 11

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.07

Analysing dot plots and stem-and-leaf plots

Date _____

Learning intentions

We are learning to:

- find the range, median, mode and mean from a dot plot or stem-and-leaf plot

Success criteria

I can:

- ☐ list or count the data values from the plot
- ☐ use the order of the plot to find the median
- ☐ calculate the mean and range

Key terms Write the definition as your teacher explains it.

Frequency

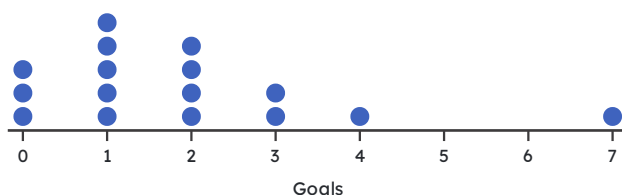
Cluster

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Set A dot plot: find the range.**Example 2** Set A dot plot: find the median.**YOU DO** On your own: Set A. Use the dot plot.

Goals scored per game



1 Range

2 Median

3 Mode

4 Mean (2 d.p.)

YOU DO On your own: Set B. Use the stem-and-leaf plot.

Test scores	
Stem	Leaf
4	5
5	0 0 1 2
6	1 3 4 6 6 6 7 8
7	4 5

Key: 5 | 2 = 52

1 Range

2 Median

3 Mode

4 Mean (2 d.p.)

Level 2 · Building up

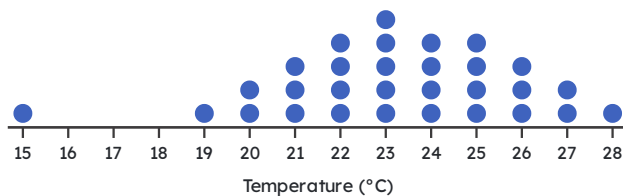
WE DO With your teacher: copy what they write on the board.

Example 3 Set B stem-and-leaf: find the mode.

Example 4 Set B stem-and-leaf: find the median.

YOU DO On your own: Set C. Use the dot plot of Bega temperatures.

Maximum temperatures at Bega in April



1 Range

2 Median

3 Mode

4 Mean (2 d.p.)

5 Mean without the outlier (2 d.p.)

6 Which measure changed most when the outlier was removed?



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Set B: find the mean (2 d.p.).
Example 6 Describe where the data in Set B clusters.
YOU DO On your own: Set D. Use the stem-and-leaf plot of player masses (ordered).

Masses of players (kg)

Stem	Leaf
7	1 4 5 6 8 9 9
8	0 0 0 2 2 3 4 5 8
9	0 0 2 2 4 8
10	4 5 6

Key: 7 | 9 = 79 kg

1 Range

2 Median

3 Mode

4 Mean (2 d.p.)

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

A dot plot has dots at 1, 2, 2, 3, 3, 3.
Find the mode.

Level 2

Find the median of 5 | 1 3 4 7 (key 5 | 1 = 51).

Level 3

Find the range of 2 | 3 8, 3 | 0 5, 4 | 1
(key 2 | 3 = 23).

Extension · Stretch yourself

Worked example

One more player of mass 120 kg joins the team.
How do the mean and median change?

1. Old mean 85.88, median 83
2. New mean 87.19, median 83.5

Answer: the mean rises more than the median

YOU DO

On your own: read the worked example, then try these.

- E1** Which measure would you use to describe the typical player's mass? Why?

- E2** Remove the 7-goal game from the dot plot. Find the new mean.

Lesson summary

Missed the lesson? Start here.

Remember

1. Count the values (n).
2. Range: highest – lowest.
3. Median: the value in position $(n + 1) \div 2$.
4. Mean: add all values (count dots $\times$ values) $\div n$.

Level 1

Goals: range

1. $7 - 0$

Answer: 7

Level 2

Goals: median

1. $n = 16$

Answer: 1.5

Level 3

Scores: mean

1. $\text{sum } 918 \div 15$

Answer: 61.2

Extension

Add a 120 kg player

1. Mean moves more

Answer: mean up more

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.08

Comparing data

Date _____

Learning intentions

We are learning to:

- compare two sets of data using plots and statistics

Success criteria

I can:

- ☐ draw a back-to-back stem-and-leaf plot
- ☐ find the mean, median, mode and range for each group
- ☐ compare the groups and decide which is more consistent

Key terms Write the definition as your teacher explains it.**Back-to-back****Consistent**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** What does the Set A plot show?**Example 2** What was the highest mark? Who scored it?**YOU DO** On your own: Set A. Use the back-to-back plot of assignment marks.

Girls	Stem	Boys
8	1	0 7
9 8	2	8 9
9 7 5	3	0 8
8	4	0
9 5 3 2 1	5	0 5 8
9 6 0	6	2 6
6 5 3	7	2 3 3 4 5
3	8	5 8
4	9	3

Key: 3 | 5 = 35

1 Highest mark? Boy or girl?

2 Lowest mark? Boy or girl?

3 How many students scored in the 70s?

4 How many girls scored 50 or more?

YOU DO On your own: Set B. Find each statistic.

1 Girls: mean

4 Boys: mean

2 Girls: median

5 Boys: median

3 Girls: range

6 Boys: range

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Find the median mark for the girls.**Example 4** Find the range for the boys.**YOU DO** On your own: Set C. Draw a back-to-back stem-and-leaf plot.

1 Class A: 45, 52, 58, 61, 63, 67, 70, 74. Class B: 38, 49, 55, 56, 60, 72, 81, 88.

Class A	Stem	Class B
	3	
	4	
	5	
	6	
	7	
	8	

Key: ____ | ____ = _____

2 For Question 1, which class did better? Use the median and range.



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Compare the girls and the boys.
Example 6 Which group is more consistent? Why?
YOU DO On your own: Set D. Compare and explain.

- 1 Comment on the differences between the girls and the boys on the assignment.

- 3 Which measure (mean or median) would you use to compare these groups? Why?

- 2 Which group was more consistent? Give reasons.

- 4 Write 2 questions you could investigate by comparing two groups at school.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What is a back-to-back stem-and-leaf plot used for?

Level 2

Group A range 12, group B range 30. Which is more consistent?

Level 3

Group A median 65, group B median 58. Which did better?

Extension · Stretch yourself

Worked example

Two basketballers' points in 5 games: Ana 12, 15, 14, 13, 16; Bo 5, 25, 8, 22, 10. Who is more consistent?

- Both have mean 14
- Ana range 4; Bo range 20

Answer: Ana (smaller range)

YOU DO

On your own: read the worked example, then try these.

- E1** Who would you pick for a game where you need at least 12 points? Why?

- E2** Make up two data sets with the same mean but very different ranges.

Lesson summary

Missed the lesson? Start here.

- Remember**
- Put the two groups on the two sides of the stems.
 - Leaves get larger moving away from the stem.
 - Compare centre: mean or median.
 - Compare spread: a smaller range means more consistent.

Level 1

Highest mark

- Largest leaf

Answer: 94, a girl

Level 2

Girls median

- Mean of the 10th and 11th

Answer: 54

Level 3

Boys range

- $93 - 10$

Answer: 83

Extension

Ana vs Bo

- Same mean, compare range

Answer: Ana

This space is left blank on purpose. It is the back of your exit ticket.